

network element responds. For example, all exchanges should respond with exactly the same message if subscriber 1 is busy.

Much work remains to standardize the network management functions of the present systems in the public telecommunications network and new systems require their own standards for network management. However, the Internet community has achieved detailed MIB specifications for many LAN and Internet systems. This has made many NMS software tools, which can manage multivendor local networks, available for LAN environments.

In this chapter we have looked at telecommunications networks, their structure, and functionality; we also introduced network management, which network operators use to improve the performance of their networks and to maintain their network in an effective way. Telecommunications network operators who build up and maintain their network have to provide good performance service at as low an investment level as possible if they are to be competitive. Their problem is how to minimize investment but still keep customers happy. To find out where they should invest and what the bottlenecks of the network are, they continuously perform traffic engineering, which is introduced in the next section.

## 2.12 Traffic Engineering

Traffic engineering is a key issue for telecommunications network operators trying to keep customers (subscribers) happy while minimizing network investments. Nowadays, network operators have to pay more and more attention to these aspects because of increasing competition in the telecommunications services market. The capacity of the network (e.g., number of channels between exchanges, exchange sizes, number of radio channels in a cellular network) should be increased where the bottlenecks of the network are found. Therefore, the utilization of the network is continuously measured and traffic demand in the future is estimated. Then, based on these estimates, the capacity of the network can be increased before severe problems occur.

An important capacity planning method is based on theoretical analyses of capacity demand and introduction to these calculations is given next.

### 2.12.1 Grade of Service

How happy subscribers are depends on the *grade of service* (GoS, availability or quality of the service) they receive. The GoS depends on the network capacity that should meet the service demand of the customers. Here we

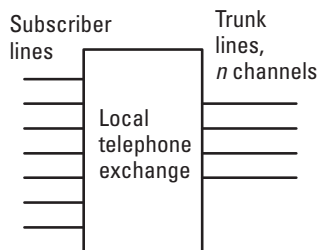
analyze only GoS for circuit-switched service and the most important factor in our study is whether the call is successful or blocked. System faults, error rates, and other quality measures are not considered here. We instead concentrate only on the evaluation of the blocking probability. In Figure 2.24 blocking occurs if more than  $n$  subscribers make external calls at a time. For the probability of unsuccessful calls, operators define the target value, the highest probability of an unsuccessful call that they assume to be acceptable for their customers. The smaller this probability is, the more capacity they have to build into the network.

Another factor we could use to define GoS is how long the subscriber has to wait until the service becomes available. We could design the network to keep customers in a queue until, for example, a transmission channel becomes free. This factor is also essential to those who plan the telephone service where a person answers incoming calls (e.g., switchboard service of an enterprise, customer service telephone).

### 2.12.2 Busy Hour

Network capacity planning is based on the so-called busy hour traffic intensity, and at other times the GoS is typically much better. *Busy hour* is an hour in the year when the average traffic intensity gets the highest value. To be accurate, the busy hour is determined by first selecting the 10 working days in a year with the highest traffic intensity; four consecutive 15-minute periods (of those 10 days) with the highest traffic intensity make up the busy hour.

The basic goal is to find a minimum capacity that gives the defined grade of service. Figure 2.24 shows a local exchange with a number of subscribers and a much smaller number  $n$  of trunk lines to the next exchange. If more than  $n$  subscribers make an external call at a time, some of them are



**Figure 2.24** Local exchange and blocking.

blocked and they have to try again. The number of external calls varies in a random manner and to be sure that blocking never occurs  $n$  should be equal to the number of subscribers. This is a far too expensive solution because the number of subscribers connected to a local exchange is usually very large and on average only a small portion of them place external calls at the same time. The principle of how to find the capacity, that is, the number of lines  $n$  in our example, that is economically feasible but acceptable from subscribers' points of view is explained next.

### 2.12.3 Traffic Intensity and the Erlang

The measure of traffic intensity for circuit-switched connections is called the *erlang* in honor of the Danish mathematician A. K. Erlang, the founder of traffic theory. The erlang unit is defined as (1) a unit of telephone traffic specifying the percentage of average use of a line or circuit (one channel) or (2) the ratio of time during which a circuit is occupied and the time for which the circuit is available to be occupied. Traffic that occupies a circuit for 1 hour during a busy hour is equal to 1 erlang. Consider these examples:

- If the traffic intensity of a subscriber line is 1 erlang, the line is occupied for 60 minutes in an hour.
- If a subscriber line is in use 6 minutes out of an hour (on average), the traffic intensity is 6 minutes/60 minutes or 100 mErl.
- The maximum traffic intensity of a 2-Mbps (30 PCM channels) line system is 30 erlangs, that is, all channels are in use 60 minutes during the busy hour.

The typical average busy-hour traffic volume generated by one subscriber is in the range of 10 to 200 mErl. Low values are typical for residential use and high values for business subscribers.

### 2.12.4 Probability of Blocking

The problem in traffic engineering is determining the capacity if the average offered traffic intensity is known (or estimated). The term *offered traffic* refers to the average generated total traffic including the traffic that is blocked in the system. Clearly the capacity should (at least usually) be higher than offered traffic; otherwise, many users would not be able to get service because all lines would be occupied all the time (on average). If all trunk lines are occupied, new users are blocked, they receive a busy tone, and they have to

try again. The essential question is this: How much higher should the capacity be for the subscribers to feel that the grade of service is acceptable?

The starting point is how often subscribers are allowed to be blocked and receive a busy tone. This probability of blockage for an acceptable GoS is usually set to be in the range of 0.2% to 5%, which means that every 500th to 20th call is blocked during a busy hour. When the average traffic load is estimated to increase to a certain volume, the network operator should increase the network capacity to keep the blocking probability below the defined GoS level.

The Poisson distribution is used as a probability model for these calculations and it gives a probability for occurrence of  $x$  events when the average number of events is  $A$  according to this formula:

$$P(x) = \frac{A^x e^{-A}}{x!} \quad (2.1)$$

where  $e = 2.71828$  and  $x!$  is the factorial of  $x$ ,  $1 \cdot 2 \cdot 3 \dots \cdot x$ . Now the average number of occupied channels is  $A$  in erlangs and (2.1) gives the probability that  $x$  number of channels is occupied at a time when a subscriber makes a call. Blocking occurs if all  $n$  channels are occupied or there may even be a need for a larger number of channels. This probability is given by:

$$P(x \geq n) = P(n) + P(n+1) + P(n+2) + \dots \quad (2.2)$$

On the other hand, one number of channels is always in use, giving this probability for

$$P(0) + P(1) + P(2) + \dots + P(n) + P(n+1) + \dots = P(x < n) + P(x \geq n) = 1 \quad (2.3)$$

and we change (2.2) into this form:

$$P(x \geq n) = 1 - P(x < n) \quad (2.4)$$

Substituting (2.1) gives

$$\begin{aligned} P(x \geq n) &= 1 - [P(0) + P(1) + \dots + P(n-1)] \\ &= 1 - \left[ \frac{A^0 e^{-A}}{0!} + \frac{A^1 e^{-A}}{1!} + \dots + \frac{A^{n-1} e^{-A}}{(n-1)!} \right] \end{aligned} \quad (2.5)$$

Now we have the Poisson formula, which is also known as the *Molina lost calls held trunking formula*, for blocking probability and it is as follows:

$$P(x \geq n) = 1 - \sum_{x=0}^{n-1} \frac{A^x e^{-A}}{x!} \quad (2.6)$$

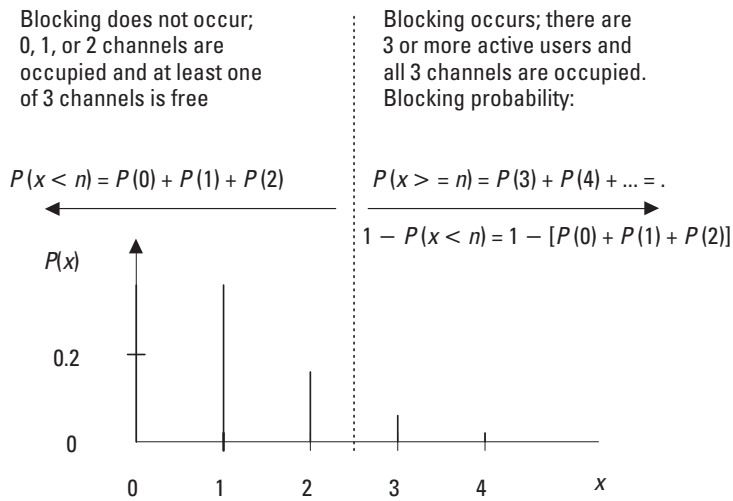
To determine the grade of service, blocking probability, we compute (using the Poisson distribution) the probability that not one channel is free when a subscriber makes a call. For this we take a value of the average (total) offered traffic as  $A$  and calculate the probability that traffic occupies all  $n$  channels or is even higher at that point in time. (The offered traffic load may be higher than  $n$  even though actual traffic can never exceed  $n$ .) We get this by subtracting from 1 the probability that traffic is smaller than  $n$  according to (2.6).

Figure 2.25 illustrates the procedure we just carried out and shows an example where average offered traffic intensity  $A = 1$  erlangs and the number of available channels  $n = 3$ . The probability density function  $P(x)$  in the figure tells the probability for each value of  $x$ , that is, the number of occupied channels. The probability that all channels are free ( $x = 0$ ) is  $P(0) = 0.37$ , one channel is occupied is  $P(1) = 0.37$ , and two channels are occupied (and one is free) is  $P(2) = 0.18$ . We subtract the sum of these probabilities from 1 and get the blocking probability, that is, the probability that the number of occupied channels  $x \geq 3$ . We get the result, when there are three channels available and average offered traffic is 1 Erl (i.e., on average, one call on all the time), that the blocking probability is 8%. This means that every twelfth call the user makes is blocked and a busy signal is received. Note that this blocking rate allows that only one channel is in use, two channels are free, and only one-third of the capacity can be utilized on average.

Equation (2.6) is based on following assumptions [2]:

- Poisson arrival rate; Poisson-distributed call attempts;
- Equal traffic volume per source;
- Lost calls held; calls that are blocked stay in the system and wait for the free channel (subscriber dials and dials again) [3];
- Infinite number of sources; if some sources are blocked or making a call, this does not affect the total offered traffic.

Let us consider another example: Total offered average traffic during the busy hour is 2 Erl ( $A = 2$ ); and the number of servers, for example,



**Figure 2.25** Probability of blocking.

transmission lines, is 5 ( $n = 5$ ). Then the probability of blockage is 5.3 % [ $P(x \geq 5) = 0.053$ ]. This means that, on average, during the busy hour every nineteenth call is blocked, a busy tone is heard, and the subscriber has to redial.

When the number of channels or servers  $n$  is high, precalculated tables like Table 2.1 are used for network planning. Such a table gives the required number of servers  $n$  when the GoS (= blocking probability in our study) and estimated offered traffic intensity  $A$  are given. For example, if the GoS is set to be 2% and offered traffic is 5 Erl (e.g., 100 subscribers with average offered traffic intensity of 50 mErl per subscriber), the network capacity should support at least 10 simultaneous calls ( $n = 10$ ). If the capacity is smaller, for example,  $n = 9$ , we get 2% with offered traffic 4.34 Erl; an offered traffic 5 Erl would give a higher blocking rate.

When we look at Table 2.1, we note that when the number of servers is small, offered traffic intensity is of the order of one-tenth of the maximum traffic intensity. For example, with two channels, offered traffic intensity at blocking probability 1% is only 150 mErl. Only one 9-minute call during an hour is allowed and both lines may be occupied on average only 4.5 minutes in an hour. The utilization of channels is less than 10%.

When the number of servers is high, allowed average traffic intensity is close to the maximum or even higher. Even when most of the channels are occupied, some channels are still free for a new call because the number of channels is large.

**Table 2.1**  
Network Capacity Planning Blocking Probability, GoS

| <i>n</i> : | 0.5% <i>A</i> | 1.0% <i>A</i> | 2.0% <i>A</i> | 3.0% <i>A</i> | 5.0% <i>A</i> | 10% <i>A</i> | 20% <i>A</i> | 50% <i>A</i> |
|------------|---------------|---------------|---------------|---------------|---------------|--------------|--------------|--------------|
| 1          | 0.01          | 0.01          | 0.02          | 0.03          | 0.05          | 0.11         | 0.25         | 1.00         |
| 2          | 0.11          | 0.15          | 0.22          | 0.28          | 0.38          | 0.60         | 1.00         | 2.73         |
| 3          | 0.35          | 0.46          | 0.60          | 0.72          | 0.90          | 1.27         | 1.93         | 4.59         |
| 4          | 0.70          | 0.87          | 1.09          | 1.26          | 1.52          | 2.05         | 2.95         | 6.50         |
| 5          | 1.13          | 1.36          | 1.66          | 1.88          | 2.22          | 2.88         | 4.01         | 8.44         |
| 6          | 1.62          | 1.91          | 2.28          | 2.54          | 2.96          | 3.76         | 5.11         | 10.4         |
| 7          | 2.16          | 2.50          | 2.94          | 3.25          | 3.74          | 4.67         | 6.23         | 12.4         |
| 8          | 2.73          | 3.13          | 3.63          | 3.99          | 4.54          | 5.60         | 7.37         | 14.3         |
| 9          | 3.33          | 3.78          | 4.34          | 4.75          | 5.37          | 6.55         | 8.53         | 16.3         |
| 10         | 3.96          | 4.46          | 5.08          | 5.53          | 6.22          | 7.51         | 9.69         | 18.3         |
| 12         | 5.28          | 5.88          | 6.61          | 7.14          | 7.95          | 9.47         | 12.0         | 22.2         |
| 15         | 7.38          | 8.11          | 9.01          | 9.65          | 10.6          | 12.5         | 15.6         | 28.2         |
| 20         | 11.1          | 12.0          | 13.2          | 14.0          | 15.3          | 17.6         | 21.6         | 38.2         |
| 25         | 15.0          | 16.1          | 17.5          | 18.5          | 20.0          | 22.8         | 27.7         | 48.1         |
| 30         | 19.0          | 20.3          | 21.9          | 23.1          | 24.8          | 28.1         | 33.8         | 58.1         |
| 35         | 23.2          | 24.6          | 26.4          | 27.7          | 29.7          | 33.4         | 40.0         | 68.1         |
| 40         | 27.4          | 29.0          | 31.0          | 32.4          | 34.6          | 38.8         | 46.2         | 78.1         |
| 45         | 31.7          | 33.4          | 35.6          | 37.2          | 39.6          | 44.2         | 52.3         | 88.1         |
| 50         | 36.0          | 37.9          | 40.3          | 41.9          | 44.5          | 49.6         | 58.5         | 98.1         |
| 55         | 40.4          | 42.4          | 44.9          | 46.7          | 49.5          | 55.0         | 64.7         | 108.1        |
| 60         | 44.8          | 46.9          | 49.6          | 51.6          | 54.6          | 60.4         | 70.9         | 118.1        |
| 65         | 49.2          | 51.5          | 54.4          | 56.4          | 59.6          | 65.8         | 77.1         | 128.1        |
| 70         | 53.7          | 56.1          | 59.1          | 61.3          | 64.7          | 71.3         | 83.3         | 138.1        |
| 75         | 58.2          | 60.7          | 63.9          | 66.2          | 69.7          | 76.7         | 89.5         | 148.1        |
| 80         | 62.7          | 65.4          | 68.7          | 71.1          | 74.8          | 82.2         | 95.8         | 158.1        |
| 85         | 67.2          | 70.0          | 73.5          | 76.0          | 79.9          | 87.7         | 102.0        | 168.0        |
| 90         | 71.8          | 74.7          | 78.3          | 80.9          | 85.0          | 93.2         | 108.2        | 178.0        |
| 95         | 76.3          | 79.4          | 83.1          | 85.9          | 90.1          | 98.6         | 114.4        | 188.0        |
| 100        | 80.9          | 84.1          | 88.0          | 90.8          | 95.2          | 104.1        | 120.6        | 198.0        |
| 110        | 90.1          | 93.5          | 97.7          | 100.7         | 105.5         | 115.1        | 133.1        | 218.0        |
| 140        | 118.0         | 122.0         | 127.0         | 130.6         | 136.4         | 148.1        | 170.5        | 278.0        |

**Table 2.1** (continued)  
Network Capacity Planning Blocking Probability, GoS

| <i>n</i> : | 0.5% <i>A</i> | 1.0% <i>A</i> | 2.0% <i>A</i> | 3.0% <i>A</i> | 5.0% <i>A</i> | 10% <i>A</i> | 20% <i>A</i> | 50% <i>A</i> |
|------------|---------------|---------------|---------------|---------------|---------------|--------------|--------------|--------------|
| 200        | 174.6         | 179.7         | 186.2         |               |               |              |              |              |
| 300        | 270.4         | 277.1         | 285.7         |               |               |              |              |              |
| 400        | 367.2         | 375.2         | 385.9         |               |               |              |              |              |
| 500        | 464.5         | 474.0         | 486.4         |               |               |              |              |              |

Note in Table 2.1 that when high blocking probability is allowed, offered traffic may be higher than the number of available channels. A part of offered traffic is blocked and actual traffic, that part which is not blocked, naturally never gets a higher value than the number of channels in erlangs.

Blockage probability can be calculated in many different ways. Table 2.1 is calculated according to erlang B formula, which assumes that blocked calls are immediately cleared and a subscriber waits and makes a new call later [3]. It gives slightly more optimistic results than the Poisson formula in (2.6). Erlang B formula is used in Europe and the Poisson formula in used in the United States for network planning. To compare the results of these two slightly different approaches, we consider an example where average offered traffic  $A = 2$  Erl. If the number of circuits  $n = 5$ , we get blocking probability  $P = 0.0367$  according to erlang B formula instead of the 0.053 we got previously with (2.6). The erlang B formula is [2] as follows:

$$P = \frac{\frac{A^n}{n!}}{\sum_{x=0}^{x=n} \frac{A^x}{x!}} \tag{2.7}$$

**2.13 Problems and Review Questions**

*Problem 2.1*

Describe how dialed digits are transferred from a subscriber’s telephone to the local exchange.

*Problem 2.2*

Explain how the telephone attenuates the speaker's voice from the microphone to the earphone. (*Hint:* Draw the current coming from the microphone in Figure 2.7 and imagine what happens to the magnetic field in the iron core of the transformer.)

*Problem 2.3*

What is a 2W/4W hybrid and why is it needed at the end of the subscriber line?

*Problem 2.4*

Explain how a 2W/4W hybrid prevents the signal from the network (receiving pair) from looping back to the transmitting pair.

*Problem 2.5*

Explain the basic principle of telephone call routing through the switching hierarchy to another region of the country.

*Problem 2.6*

A network has  $N$  subscribers. Each subscriber is connected directly to all other subscribers.

- (a) What is the total number of lines  $L$  in the network?
- (b) What is the value of  $L$  for  $N = 2, 10, 100$ , and  $1,000$ ?
- (c) How many lines must be built to each subscriber?
- (d) Is this kind of network structure suitable for a public telecommunications network? Explain.

*Problem 2.7*

What are the basic differences between the public and private telecommunications networks? List a few examples of both public and private networks.

*Problem 2.8*

What is ISDN? How does the service and structure of the subscriber interface differ from the conventional analog telephone service?

*Problem 2.9*

How does an IN differ from conventional fixed telephone network? List some examples of IN services.

**Problem 2.10**

A PBX/PABX has seven telephone channels to a public exchange. During the busy hour, on average, 3.4 lines are occupied. (a) What is the traffic intensity during the busy hour? (b) Estimate, with the help of the Table 2.1, the GoS (blocking probability).

**Problem 2.11**

What is the total offered traffic intensity from a PBX/PABX to PSTN if 10 calls are made, each with a duration of 6 minutes during 1 hour?

**Problem 2.12**

A subscriber makes one 6-minute call in one day between 10:00 and 10:06. What is the average traffic intensity of her subscriber line during (a) 10:00–10:06, (b) 10:00–10:15, (c) 10:00–11:00, and (d) 00:00–24:00 of that day?

**Problem 2.13**

Use the Poisson (or “Molina lost calls held”) trunking formula to calculate the blocking probability (GoS) when the total offered traffic is 2 Erl and the number of available transmission channels in the network is 5.

**Problem 2.14**

Draw two curves for GoS levels of 1% and 10%. Use the vertical axis as a ratio  $A/n$  from 1% to 100% and the horizontal axis as a number of circuits  $n$  from 1 to 20. Use traffic engineering Table 2.1. What can you say about network utilization when the number of circuits  $n$  is small? How does the utilization of the circuits depend on the allowed probability of blocking?

**Problem 2.15**

What will the approximate capacity of a network be (i.e., how many channels should be available) if there are 100 subscribers and each of them generates offered traffic of 40 mErl? The probability of blocking is (a) 20% and (b) 1%. Use traffic engineering Table 2.1.

**Problem 2.16**

There are 20 users of a keyphone system that has two lines to a public network. What is the blocking probability when each user generates a 100-mErl offered traffic?

**Problem 2.17**

A keyphone system with three lines to the local exchange is used in an office of 10 persons. Each of them uses the phone for an external call of 15 minutes in a busy hour. How many lines are reserved on average during an hour? What is the blocking probability? What do you think about the capacity of this system?

**Problem 2.18**

Subscribers of a local exchange generate 100 mErl of traffic through the exchange to the network. What should the number of trunk channels be if the number of subscribers in the area is (a) 10, (b) 100, (c) 1,000, and (d) 4,000? The allowed blocking level is 1%. Use Table 2.1 to estimate the required number of circuits.

## References

- [1] *Telecommunications Transmission Engineering*, Bellcore Technical Publications, 1990.
- [2] Freeman, R. L., *Telecommunication System Engineering*, 3rd ed., New York: John Wiley & Sons, 1996.
- [3] Freeman, R. L., *Fundamentals of Telecommunications*, New York: John Wiley & Sons, 1999.